

Shivani Bhandari

AI Agents | RAG Systems | FastAPI | ONNX Runtime | LLM Inference Quantization

📍 Gurugram, India | 📞 +91 7530815555 | ✉ Shivani215143@gmail.com

🌐 <https://www.linkedin.com/in/shivani-bbb69a333/> | 🐙 github.com/Shivani767

EXPERIENCE

Josh Technology Group

Software Engineer (AI/LLM Systems)

Jul 2025 – Present

Gurugram, India

- Developed AI-powered features for Pod.ai, an LLM-driven recruitment platform, enabling candidate-job matching, resume intelligence, and recruiter assistance workflows.
- Built Retrieval-Augmented Generation (RAG) pipelines using embeddings and FAISS, improving search relevance and contextual recommendations across large candidate and job datasets.
- Fine-tuned Transformer models using LoRA/QLoRA for resume summarization, ranking, and contextual question answering.
- Designed and deployed FastAPI-based inference services integrated with production microservices for scalable AI workflows.
- Optimized model serving with ONNX Runtime, quantization, and batching techniques to improve latency and throughput of real-time AI applications.
- Collaborated with product, engineering, and data teams to design evaluation frameworks, monitor model performance, and run experimentation pipelines.

Pisoft Informatics Pvt. Ltd.

Python Developer Intern (AI/ML)

Jan 2025 – Jun 2025

Mohali, India

- Built AI-powered ERP modules for document intelligence, invoice processing, workflow automation, and enterprise search applications.
- Developed a FAISS-based semantic search engine over financial and operational documents, reducing document retrieval time by 35%.
- Implemented Transformer-based summarization and contextual query systems, improving information retrieval accuracy by 28%.
- Designed data ingestion and preprocessing pipelines for structured and unstructured enterprise datasets using Python and Pandas.
- Deployed ONNX-optimized models on Azure ML, reducing inference latency by 22% and enabling scalable API-based model serving.

Design Innovation Centre (DIC), Panjab University

Machine Learning Intern (MDaRT)

Mar 2023 – Dec 2023

Chandigarh, India

- Developed an RCNN-based dental image classification system on 1,000+ clinical images, improving automated furcation detection accuracy through feature engineering, data preprocessing, and hyperparameter optimization; deployed the solution using Flask and React for clinical decision support.
- **Project Monitoring Dashboard** [Link] — Designed and deployed a Looker Studio dashboard tracking 50+ research projects, enabling real-time KPI visualization, progress monitoring, and stakeholder reporting.

PROJECTS

InferLite – LLM Inference & Quantization Research Platform | Python, FastAPI, ONNX Runtime, PyTorch

- Built a research-grade platform for benchmarking 4+ LLM runtimes (vLLM, ONNX Runtime, TensorRT-LLM, llama.cpp) and evaluating 5+ quantization methods including GPTQ, AWQ, INT8, FP16, and GGUF.
- Developed automated pipelines for TTFT, tokens/sec, P95 latency, throughput, memory, and cost analysis, achieving up to 70% memory reduction and 2.5× higher inference throughput through quantization-driven optimization.
- Implemented ONNX graph optimization, distributed inference simulation, GPU telemetry monitoring, and hardware-aware deployment recommendations for large-scale LLM serving experiments.

LangGraph Open Source Contributions | Python, Distributed Systems, Agent Frameworks

- Contributed to LangGraph's Pregel runtime by improving DeltaChannel state persistence, checkpoint reconstruction, and update-state behavior across distributed agent workflows.
- Proposed Human-in-the-Loop workflow abstractions and participated in ApprovalNode design discussions for production-grade agent orchestration and long-running approval processes.

AutoGen Multi-Agent Systems | AutoGen, LLM Agents, FastAPI

- Built multi-agent workflows comprising 5+ specialized agents for planning, reasoning, tool execution, and autonomous task coordination.
- Implemented memory management, agent communication, and orchestration pipelines, reducing manual intervention by 60%+ in complex multi-step automation workflows.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL

AI/ML: Machine Learning, Deep Learning, Transformers, LoRA/QLoRA, Computer Vision, Model Evaluation

LLMs Agentic AI: RAG, LangGraph, AutoGen, AI Agents, Embeddings, Vector Search, Function Calling

LLM Systems: ONNX Runtime, vLLM, TensorRT-LLM, llama.cpp, GPTQ, AWQ, GGUF, Quantization, Inference Optimization

Backend: FastAPI, Flask, REST APIs, Microservices, Async Programming

Databases: PostgreSQL, MySQL, Redis, FAISS

Cloud DevOps: AWS, Docker, Git, GitHub Actions, Linux, CI/CD

Monitoring: Prometheus, Grafana, OpenTelemetry, MLflow, Jupyter

AWARDS AND CERTIFICATIONS

Best Employee of the Quarter – Josh Technology Group

CodeChef 6-Star Programmer – Rating 2464+, Top 0.2

LeetCode – Rating 1700+, 500+ Problems Solved

Microsoft Certified: Responsible Generative AI

Microsoft Learn Cloud Skills Challenge

GSSoC Contributor – Earned Prestigious Points Badge

EDUCATION

UIET, Panjab University <i>B.E. in Electronics and Communication Engineering (CGPA: 8.09/10)</i>	2021 – 2025 Chandigarh, India
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